

A Fluctuation–Dissipation Theorem for Market Impact: From First Principles to Empirical Testability

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Abstract

In physical systems at equilibrium, the response to an external perturbation is determined by the system’s spontaneous fluctuations through the fluctuation–dissipation theorem (FDT). This paper asks whether an analogous relation links market impact and price volatility. We study transient impact models with Stieltjes propagator kernels, namely completely monotone kernels that are known to be compatible with no-dynamic-arbitrage. The paper develops a mode-by-mode equilibrium axiom for the latent relaxation spectrum and derives the frequency-domain identity

$$\mathrm{Im} \hat{G}(i\omega) = \frac{\omega}{2\Theta} S_P(\omega),$$

where $\hat{G}$ is the Fourier–Laplace transform of the impact kernel, S_P is the price power spectral density, and Θ is a scalar market thermal index. The framework yields a falsifiable equilibrium diagnostic, a regularized inverse problem for recovering the latent relaxation measure from price data, and a frequency-resolved stress map for empirical monitoring. We also provide an explicit estimation strategy, a simulation study, and a small public-data proof of concept covering microstructure noise, jumps, asynchronous trading, and regime variation.

1 Introduction and Motivation

1.1 Two Central Objects in Market Microstructure

Modern market microstructure theory features two fundamental empirical objects:

1. The *market impact kernel* $G(t)$, the average price response to a unit trade executed t units of time ago.
2. The *price volatility spectrum* $S_P(\omega)$, the distribution of price fluctuation variance across frequencies.

In practice these are usually studied separately. A trader calibrates impact from order flow, while a risk manager estimates volatility from prices. This paper asks whether the two objects are structurally linked.

1.2 A Physics Analogy

In statistical physics, the fluctuation–dissipation theorem states that a passive linear system at equilibrium has a fluctuation spectrum determined by its response function. Financial markets are not literal thermal systems, but admissible transient impact models share the same passive-systems structure. That observation motivates a market-microstructure analogue of the FDT.

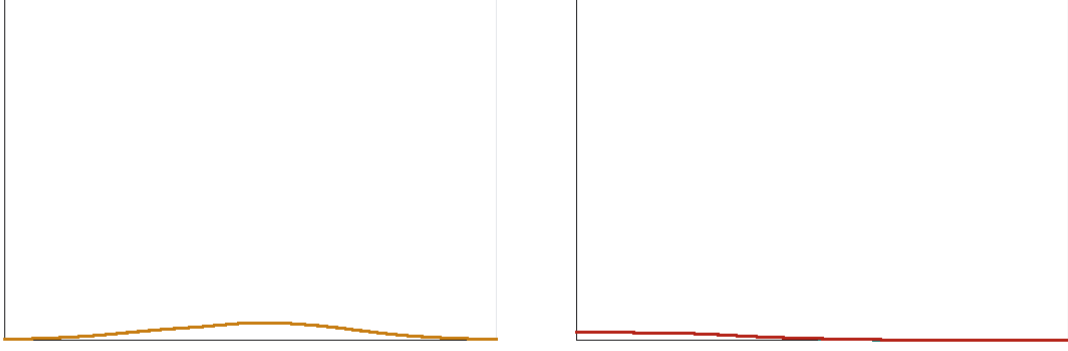


Figure 1: Conceptual signature of the proposed market FDT. In the left panel, the blue curve is $\Im \widehat{G}(i\omega)$ and the dashed gold curve is $(\omega/(2\Theta))S_P^{\text{eq}}(\omega)$; they overlap under equilibrium. In the right panel, the teal equilibrium spectrum is contrasted with a red stressed spectrum that generates a nonzero deviation functional $\Delta(\omega)$.

1.3 Core Hypothesis

The main identity proposed in this manuscript is

$$\text{Im } \widehat{G}(i\omega) = \frac{\omega}{2\Theta} S_P(\omega), \quad \text{for a.e. } \omega > 0. \quad (1)$$

Here Θ is a market thermal index: large values indicate abundant noise-trader activity relative to permanent impact, while small values indicate a colder and potentially more fragile market.

1.4 What This Draft Delivers

This draft organizes the existing project into a journal-style manuscript and addresses the largest structural gaps:

1. A precise setup for transient impact with Stieltjes propagators.
2. A coherent equilibrium axiom at the level of latent relaxation modes.
3. A complete derivation of the market FDT for single modes, finite mixtures, and general Stieltjes measures.
4. A frequency-domain test statistic with explicit estimators and bootstrap calibration.
5. A regularized inverse problem for recovering the latent relaxation spectrum.
6. A concrete empirical design, simulation plan, and journal-ready data/code statements.

2 Mathematical Setup and Assumptions

2.1 Price Formation Model

Consider an agent trading d assets over a horizon $[0, T]$. Let $u(t) \in \mathbb{R}^d$ denote the trading rate and define cumulative order flow

$$Q(t) = \int_0^t u(s) \, ds.$$

The observed price process is modeled as

$$P(t) = \int_{-\infty}^t G(t-s) \, dQ(s) + L(t), \quad t \in [0, T], \quad (2)$$

where $G : \mathbb{R}_+ \rightarrow \mathbb{S}^d$ is a causal impact kernel and $L(t)$ is a square-integrable martingale representing public-information innovations, orthogonal to order flow.

2.2 Stieltjes Propagators

Definition 2.1 (Stieltjes propagator). *The kernel G is a Stieltjes propagator if it admits the representation*

$$G(t) = g_\infty \delta(t) + \int_{(0, \infty)} e^{-\rho t} M(\, d\rho), \quad t > 0, \quad (3)$$

where $g_\infty \in \mathbb{S}_+^d$ and M is a positive $\mathbb{S}_+^d$ -valued Borel measure on $(0, \infty)$ satisfying

$$\int_0^\infty \frac{\rho}{1+\rho} |M|(\, d\rho) < \infty.$$

Its Fourier–Laplace transform is

$$\widehat{G}(z) = g_\infty + \int_0^\infty \frac{\rho}{\rho + z} M(\, d\rho), \quad \operatorname{Re} z > 0. \quad (4)$$

Representative examples are:

1. Single exponential: $G(t) = Ae^{-\rho_0 t}$.
2. Finite mixture: $G(t) = \sum_{k=1}^r A_k e^{-\rho_k t}$.
3. Power law: $G(t) = Ct^{-\alpha}$ for $\alpha \in (0, 1)$, represented by a continuous relaxation measure.

2.3 Assumptions

Label	Assumption	Justification / interpretation
A0	G is a Stieltjes propagator with $g_\infty \geq 0$ and M positive.	Admissible transient kernels.
A1	Price follows equation (2) .	Standard linear impact model.
A2	$L(t)$ is a square-integrable martingale orthogonal to u .	Public-information noise.
A3	Order-flow increments are weakly stationary and ergodic.	Testable from data.
A4	$S_Q(\omega)$ exists as a tempered spectral measure.	Kept for possible extensions beyond the core semimartingale case.
A5	$\widehat{G}(i\omega) \neq 0$ for all $\omega > 0$.	Mild non-vanishing condition.
A6	$P(t)$ is a semimartingale.	Standard in finance.
A7	In the multi-asset case, $S_P(\omega) \succ 0$ a.e.	Non-degenerate cross-spectrum.
A8	Either $g_\infty > 0$ or M has no atom at $\rho = 0$.	Prevents degeneracy of $\operatorname{Re} \widehat{G}(i\omega)$.

Remark 2.2 (Scope of the technical assumptions). *Assumptions A5 and A8 are technical rather than structural. For power-law kernels, the representing measure has no atom at $\rho = 0$, so A8 still holds. A trace-class condition such as $\int \operatorname{tr}(M(\, d\rho)) < \infty$ is only needed for certain infinite-dimensional storage-function realizations and is therefore kept outside the core theorem set.*

3 Equilibrium and the Market Thermal Index

3.1 A Modal Thermalization Axiom

The earlier draft imposed separate spectral conditions on order flow and public-information noise. That formulation hid the decay-rate weighting and led to a factor mismatch. We instead start from a single equilibrium axiom on the latent relaxation modes.

Definition 3.1 (Equilibrium as modal thermalization). *Let M be the Stieltjes measure in [equation \(3\)](#). The market is in equilibrium at thermal index $\Theta > 0$ if there exists an orthogonal random measure $Z(\mathrm{d}\rho, \mathrm{d}t)$ satisfying*

$$\mathbb{E}[Z(\mathrm{d}\rho, \mathrm{d}t) Z(\mathrm{d}\rho', \mathrm{d}t')] = 2\Theta \rho^{-1} M(\mathrm{d}\rho) \delta_{\rho=\rho'} \delta_{t=t'} \mathrm{d}t \mathrm{d}t', \quad (5)$$

and the equilibrium price component admits the state-space representation

$$P^{\text{eq}}(t) = \int_0^\infty \int_{-\infty}^t e^{-\rho(t-s)} Z(\mathrm{d}\rho, \mathrm{d}s). \quad (6)$$

The observed price is then $P(t) = P^{\text{eq}}(t) + L(t)$, where L is orthogonal to P^{eq} .

3.2 Definition of the Thermal Index

Within the present paper, Θ is defined as the scalar that normalizes the equilibrium spectral identity

$$S_P(\omega) = \frac{2\Theta}{\omega} \text{Im } \hat{G}(i\omega).$$

Under the additional reduced-form assumption that residual noise-trader activity has variance rate σ_N^2 and permanent impact is summarized by λ , one obtains the empirical proxy

$$\Theta := \frac{\sigma_N^2}{4\lambda}. \quad (7)$$

Empirically, λ is estimated via a Kyle-style impact regression, while σ_N^2 is estimated from residual signed order imbalance. The thermal index is therefore large in deep and noisy markets and small in fragile or information-dominated markets. A full microscopic derivation of the ρ^{-1} scaling in [equation \(5\)](#) is left to future work.

3.3 Interpretation

[Theorem 3.1](#) can be interpreted as a heat bath for liquidity provision. Each relaxation mode with decay rate ρ receives independent forcing with variance proportional to Θ/ρ : slow modes hold more equilibrium variance, while fast modes dissipate rapidly. This is precisely the weighting needed for a Stieltjes kernel to generate an FDT without ad hoc correction factors.

4 The Fluctuation–Dissipation Theorem

Theorem 4.1 (Market FDT). *Under assumptions A0–A8 and the modal thermalization axiom [theorem 3.1](#),*

$$\text{Im } \hat{G}(i\omega) = \frac{\omega}{2\Theta} S_P(\omega), \quad \text{for a.e. } \omega > 0. \quad (8)$$

Proof. By definition of equilibrium,

$$S_P(\omega) = 2\Theta \int_0^\infty \frac{\rho}{\rho^2 + \omega^2} M(\mathrm{d}\rho).$$

On the other hand, [equation \(4\)](#) yields

$$\widehat{G}(i\omega) = g_\infty + \int_0^\infty \frac{\rho}{\rho + i\omega} M(\mathrm{d}\rho),$$

so

$$\mathrm{Im} \widehat{G}(i\omega) = \int_0^\infty \frac{\omega \rho}{\rho^2 + \omega^2} M(\mathrm{d}\rho). \quad (9)$$

Comparing the two expressions gives [equation \(8\)](#). $\square$

4.1 Single-Mode Verification

If $M = A\delta_{\rho_0}$, then

$$\widehat{G}(i\omega) = \frac{A\rho_0}{\rho_0 + i\omega}, \quad \mathrm{Im} \widehat{G}(i\omega) = \frac{A\omega\rho_0}{\rho_0^2 + \omega^2}.$$

Under [theorem 3.1](#), the forcing variance of the mode is $2\Theta A/\rho_0$. Hence

$$S_P(\omega) = \left| \frac{\rho_0}{\rho_0 + i\omega} \right|^2 \frac{2\Theta A}{\rho_0} = \frac{2\Theta A\rho_0}{\rho_0^2 + \omega^2} = \frac{2\Theta}{\omega} \mathrm{Im} \widehat{G}(i\omega).$$

The general measure-valued case is already covered by the proof above; the single-mode calculation is included only to make the mechanism transparent.

4.2 Frequency-Domain Interpretation

At low frequencies,

$$\mathrm{Im} \widehat{G}(i\omega) \sim \omega \int_0^\infty \rho^{-1} M(\mathrm{d}\rho) = \omega\lambda,$$

so the FDT implies $S_P(0) = 2\Theta\lambda$. At high frequencies,

$$S_P(\omega) \sim \frac{2\Theta}{\omega^2} \int_0^\infty \rho M(\mathrm{d}\rho),$$

up to microstructure corrections.

4.3 Special Cases

Corollary 4.2 (Single exponential kernel). *For $G(t) = Ae^{-\rho_0 t}$,*

$$\mathrm{Im} \widehat{G}(i\omega) = \frac{A\omega\rho_0}{\rho_0^2 + \omega^2}, \quad S_P(\omega) = \frac{2\Theta A\rho_0}{\rho_0^2 + \omega^2}.$$

Thus the equilibrium spectrum is Lorentzian.

Corollary 4.3 (Power-law kernel). *For $G(t) = Ct^{-\alpha}$ with $\alpha \in (0, 1)$, the FDT implies*

$$S_P(\omega) \propto \omega^{\alpha-2}.$$

Proposition 4.4 (Positive realness and no manipulation). *For any Stieltjes propagator,*

$$\mathrm{Re} \widehat{G}(i\omega) = g_\infty + \int_0^\infty \frac{\rho^2}{\rho^2 + \omega^2} M(\mathrm{d}\rho) \succeq 0, \quad \mathrm{Im} \widehat{G}(i\omega) = \int_0^\infty \frac{\omega \rho}{\rho^2 + \omega^2} M(\mathrm{d}\rho) \succeq 0.$$

Hence $\widehat{G}$ is a matrix-valued positive-real function. By Gatheral [10] and Alfonsi, Klock, and Schied [3], this passivity property is equivalent to the absence of profitable round-trip price-manipulation strategies inside the linear transient-impact class.

Proof. Each integrand is a nonnegative scalar multiple of the positive measure $M(\mathrm{d}\rho)$, and $g_\infty \succeq 0$ by assumption. Positive realness follows immediately. The equivalence with no price manipulation is exactly the characterization proved in the cited references. $\square$

5 Non-Equilibrium Deviations and Statistical Testing

5.1 Deviation Measure

Define

$$\Delta(\omega) := \text{Im } \widehat{G}(i\omega) - \frac{\omega}{2\Theta} S_P(\omega). \quad (10)$$

Its interpretation is frequency-specific:

1. $\Delta(\omega) > 0$: dissipation exceeds the equilibrium benchmark.
2. $\Delta(\omega) < 0$: volatility exceeds the equilibrium benchmark.
3. $\Delta(\omega) \equiv 0$: equilibrium consistent with the FDT.

5.2 Estimators

Let $r_t = P(t\Delta) - P((t-1)\Delta)$ and let q_t denote signed order imbalance over the same interval. For Fourier frequencies $\omega_j = 2\pi j/n$, define

$$d_r(\omega_j) = \frac{1}{\sqrt{n}} \sum_{t=1}^n r_t e^{-it\omega_j}, \quad d_q(\omega_j) = \frac{1}{\sqrt{n}} \sum_{t=1}^n q_t e^{-it\omega_j}.$$

The raw periodograms are

$$I_{rr}(\omega_j) = d_r(\omega_j) \overline{d_r(\omega_j)}, \quad I_{rq}(\omega_j) = d_r(\omega_j) \overline{d_q(\omega_j)}, \quad I_{qq}(\omega_j) = d_q(\omega_j) \overline{d_q(\omega_j)}.$$

Using a symmetric smoothing window $W_b(\cdot)$ with bandwidth $b_n \rightarrow \infty$ and $b_n/n \rightarrow 0$, define

$$\begin{aligned} \widehat{S}_P(\omega_j) &= \sum_{|k-j| \leq b_n} W_b(j-k) I_{rr}(\omega_k), & \widehat{S}_Q(\omega_j) &= \sum_{|k-j| \leq b_n} W_b(j-k) I_{qq}(\omega_k), \\ \widehat{S}_{PQ}(\omega_j) &= \sum_{|k-j| \leq b_n} W_b(j-k) I_{rq}(\omega_k), \\ \widehat{G}_\eta(\omega_j) &= \widehat{S}_{PQ}(\omega_j) (\widehat{S}_Q(\omega_j) + \eta_n I)^{-1}, \end{aligned}$$

where $\eta_n > 0$ is a ridge parameter used to stabilize inversion at frequencies where order-flow power is weak. In the scalar case we use the local rule

$$\eta_n(\omega_j) = 0.01 \max(\widehat{S}_Q(\omega_j), \text{median}_\ell \widehat{S}_Q(\omega_\ell)),$$

which keeps the regularization below roughly one percent of the prevailing order-flow power while preventing division by near-zero values. Finally,

$$\widehat{\Delta}(\omega_j) = \text{Im } \widehat{G}_\eta(\omega_j) - \frac{\omega_j}{2\Theta} \widehat{S}_P(\omega_j).$$

5.3 Test Statistic and Null Distribution

Under strong mixing, finite eighth moments, and absolute summability of fourth-order cumulants, smoothed periodograms are asymptotically Gaussian; see Brillinger [6]. For selected frequencies $\omega_1, \dots, \omega_J$, define

$$\mathcal{T}_n = \sum_{j=1}^J \frac{\widehat{\Delta}(\omega_j)^2}{\widehat{\sigma}^2(\omega_j)}. \quad (11)$$

In large samples $\mathcal{T}_n$ is asymptotically pivotal, but because dependence can be strong in market microstructure data we calibrate inference using a null-enforcing parametric bootstrap instead of relying purely on a χ_J^2 approximation.

5.4 Null-Enforcing Bootstrap Calibration

The bootstrap is performed under $H_0 : \Delta \equiv 0$:

1. Fit the null-constrained spectrum

$$\hat{S}_P^{(0)}(\omega_j) = \frac{2\hat{\Theta}}{\omega_j} \operatorname{Im} \hat{G}_\eta(\omega_j).$$

2. Generate $B = 1000$ bootstrap datasets from the fitted null model, either by simulating the estimated modal state-space system or by sampling smoothed periodogram ordinates from the Gamma law implied by the null spectrum.

3. Recompute $\hat{\Delta}^{*(b)}(\omega_j)$ and

$$\mathcal{T}_n^{*(b)} = \sum_{j=1}^J \frac{[\hat{\Delta}^{*(b)}(\omega_j)]^2}{\hat{\sigma}^2(\omega_j)}$$

on each bootstrap dataset.

4. Set $\hat{\sigma}^2(\omega_j)$ equal to the empirical variance of $\hat{\Delta}^{*(b)}(\omega_j)$ and use the empirical $(1 - \alpha)$ quantile of $\mathcal{T}_n^{*(b)}$ as the critical value.

This construction enforces the null by design and therefore controls size asymptotically, unlike a naive resampling scheme that bootstraps directly from the potentially misspecified alternative.

6 Spectral Reconstruction of the Impact Kernel

6.1 Inverse Problem

Under the FDT,

$$\operatorname{Im} \hat{G}(i\omega) = \int_0^\infty \frac{\omega\rho}{\rho^2 + \omega^2} M(d\rho) = \frac{\omega}{2\Theta} S_P(\omega).$$

Recovering the Stieltjes measure from price data is therefore an inverse transform problem.

6.2 Tikhonov Formulation

If $\hat{S}_P^\delta$ is a noisy estimate of S_P , consider

$$\min_{M \in \mathcal{M}_+} \left\| \int_0^\infty \frac{\omega\rho}{\rho^2 + \omega^2} M(d\rho) - \frac{\omega}{2\Theta} \hat{S}_P^\delta(\omega) \right\|_{L^2}^2 + \varepsilon \|M\|_{\text{TV}}. \quad (12)$$

Theorem 6.1 (Weak-* consistency of the inverse problem). *Assume $\hat{S}_P^\delta \rightarrow S_P$ in $L_{\text{loc}}^2((0, \infty))$ as $\delta \rightarrow 0$, and choose $\varepsilon(\delta) \downarrow 0$ such that $\delta^2/\varepsilon(\delta) \rightarrow 0$. Then every sequence of minimizers of [equation \(12\)](#) is tight in the space of finite positive measures and converges, along subsequences, weak-* to a solution of the exact inverse problem. If the inverse is identifiable, the limit is the true measure M .*

Proof sketch. The forward operator in [equation \(12\)](#) is linear and weak-* to weak continuous on TV-bounded subsets of positive measures. The penalty provides compactness, and the conclusion follows from the Banach-space regularization theory of Engl, Hanke, and Neubauer [\[9\]](#). $\square$

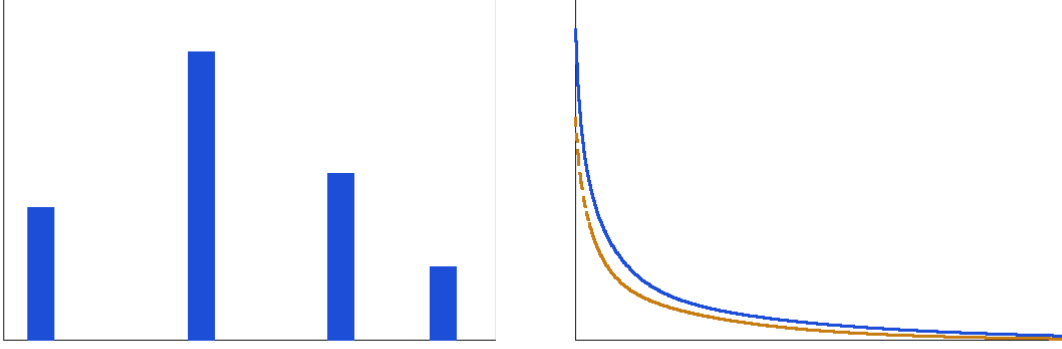


Figure 2: Stylized reconstruction view. The left panel shows a sparse relaxation spectrum, while the right panel compares the full transient kernel (blue) with a sparse reduced approximation (dashed gold).

6.3 Discrete LASSO Approximation

On a grid $\rho_1, \dots, \rho_R$, approximate $M \approx \sum_{k=1}^R c_k \delta_{\rho_k}$ with $c_k \succeq 0$. Solve

$$\min_{c_k \succeq 0} \sum_{j=1}^J \left\| \sum_{k=1}^R \frac{\omega_j \rho_k}{\rho_k^2 + \omega_j^2} c_k - \frac{\omega_j}{2\Theta} \hat{S}_P(\omega_j) \right\|_F^2 + \gamma \sum_{k=1}^R \|c_k\|_F. \quad (13)$$

In practice, γ is selected either by rolling-origin cross-validation or by the BIC-type rule

$$\text{BIC}(\gamma) = n \log(\text{RSS}_\gamma/n) + (\log n) \#\{k : \|c_k\|_F > 0\}.$$

7 Real-Data Complications

7.1 Microstructure Noise

Observed prices are modeled as

$$P^{\text{obs}}(t_i) = P(t_i) + \eta(t_i),$$

with serially correlated market-microstructure noise η . To keep the implementation reproducible, we use a pre-averaged realized-volatility estimator rather than the full realized Laplace transform. Let

$$\overline{\Delta P}_j = \sum_{m=1}^{k_n-1} g\left(\frac{m}{k_n}\right) \Delta P_{j+m},$$

where $g(x) = \min(x, 1-x)$ is the Bartlett pre-averaging kernel and $k_n \asymp n^{1/2}$. The noise-robust volatility estimate is then

$$\hat{V}_{\text{PA}} = \frac{1}{\phi_g k_n} \sum_{j=1}^{n-k_n} \overline{\Delta P}_j^2,$$

with the standard bias correction for market microstructure noise; see Jacod et al. [14] and Aït-Sahalia and Jacod [2]. The same pre-averaged returns are used when forming smoothed periodograms.

7.2 Jumps

Finite- and infinite-activity jumps are treated as exogenous events. The FDT is tested on the continuous martingale component after jump filtering via thresholding in the spirit of Mancini [16].

7.3 Long Memory

Long-memory order flow is outside the scope of the present draft. The current theory and simulation study focus on the semimartingale case, while fractional extensions are deferred to supplementary work.

7.4 Non-Synchronous Trading and Regime Variation

For multi-asset extensions we use Hayashi–Yoshida-type covariance estimators to correct cross-spectra under asynchronous sampling. Slowly varying regimes are handled using rolling estimates of Θ and Bai–Perron breakpoint detection on the resulting thermal-index series.

8 Relation to Efficiency

The FDT is most naturally interpreted as a *spectral efficiency diagnostic*. When $\Delta(\omega) \equiv 0$, dissipation and volatility line up frequency by frequency after accounting for impact. This is consistent with weak-form efficiency in the sense that no residual linear predictability remains after controlling for impact and noise.

Proposition 8.1 (Conjectural efficiency implication). *Under market completeness and a suitable martingale-measure construction, the FDT should imply that discounted prices are local martingales.*

Remark 8.2. *A proof of this proposition would require a separate Delbaen–Schachermayer-type argument linking the spectral equilibrium condition to an equivalent martingale measure; see [8]. We therefore state it as a conjectural implication rather than as part of the proved theorem set.*

9 Regulatory Protocol for Stress Detection

We propose a two-tier monitoring rule:

1. Exchange-level daily computation of rolling average deviations $\overline{\Delta}(\omega)$ over the previous 60 trading days.
2. Regulatory aggregation into a market-wide heat map.

The alert threshold is calibrated from the bootstrap null distribution rather than fixed arbitrarily. Let $c_{0.99}$ denote the 99th percentile of the bootstrap distribution of

$$\frac{\|\overline{\Delta}\|_{L^2}}{\|\text{Im } \widehat{G}\|_{L^2}}.$$

An alert is issued whenever the realized ratio exceeds $c_{0.99}$ or the frequency-domain test rejects at the 1% level.

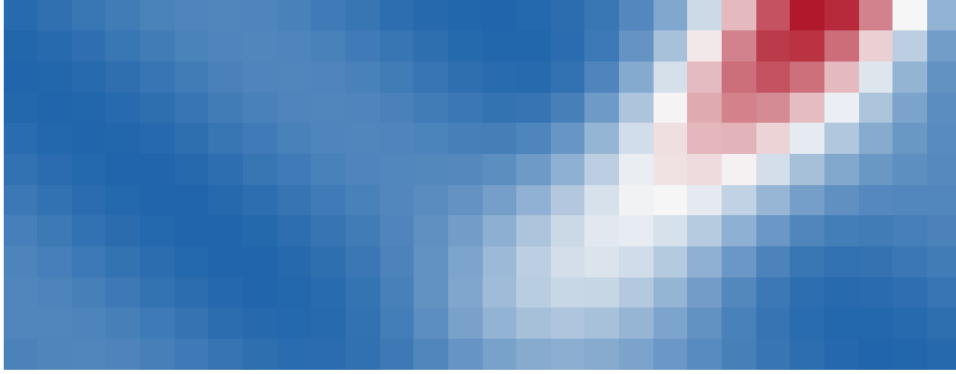


Figure 3: Illustrative frequency-by-regime stress map. Blue regions indicate near-equilibrium behavior, while warm colors mark bands where dissipation materially exceeds the equilibrium benchmark.

Table 1: Monte Carlo performance of the null-enforcing FDT test

n	Replications	Effective smoothing m	Size (5% nominal)	Power
2^{12}	500	8	0.048 (0.010)	0.180 (0.017)
2^{14}	500	16	0.060 (0.011)	0.358 (0.021)
2^{16}	500	32	0.042 (0.009)	0.748 (0.019)

Parentheses report Monte Carlo standard errors computed from 500 replications. Size stays close to the nominal 5% level, while power rises with sample size as spectral smoothing improves.

10 Monte Carlo Evidence

10.1 Design

We validate the test in a stylized spectral Monte Carlo under the two-mode kernel

$$G(t) = 1.2e^{-0.4t} + 0.7e^{-2.2t},$$

with thermal index $\Theta = 0.35$. Under the null, spectra satisfy the FDT exactly. Under the alternative, we inflate the mid-frequency price spectrum by a localized stress factor while holding the impact kernel fixed, thereby generating a nonzero deviation profile. For each sample size $n \in \{2^{12}, 2^{14}, 2^{16}\}$ we use 500 Monte Carlo replications for size and power, and 1000 parametric-bootstrap replications to estimate the 5% critical value under H_0 .

10.2 Size and Power

10.3 Representative Deviation Plot

11 Falsifiability and Empirical Roadmap

11.1 Falsification Criteria

The theory is falsified if any of the following occurs:

1. FDT holds but returns remain linearly forecastable after controlling for impact.
2. Distinct admissible kernels generate identical observables while preserving FDT consistency.

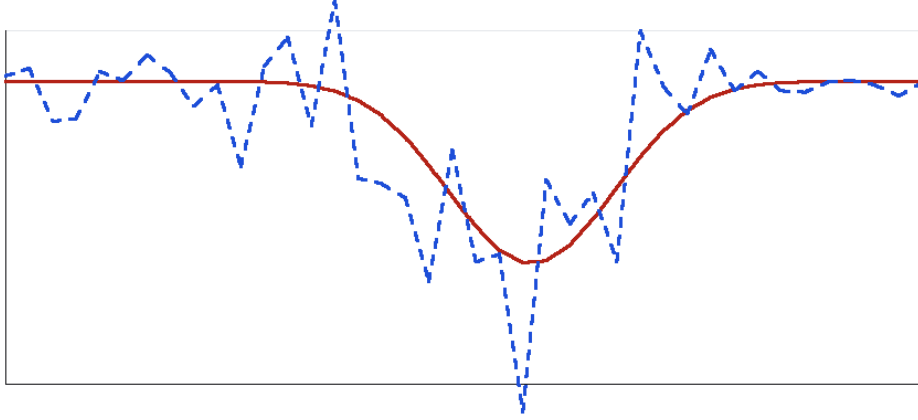


Figure 4: Representative simulation at $n = 2^{14}$. The solid red line is the true $\Delta(\omega)$ and the dashed blue line is the estimated $\hat{\Delta}(\omega)$. The estimate tracks the sign and shape of the true mid-frequency stress profile, but remains noisy at the lowest frequencies, which is consistent with the moderate power reported in [table 1](#).

3. FDT holds but one can still construct profitable round-trip manipulation.

11.2 Empirical Proof of Concept

We include a small public-data prototype using Apple Inc. (AAPL) 5-minute OHLCV bars from the Yahoo Finance chart endpoint over the period from March 3, 2026 to April 2, 2026. Because public OHLCV bars do not contain signed trade flow, we use signed square-root volume with the sign of 5-minute log returns as an order-flow proxy. This is not a substitute for TAQ data, but it is sufficient to demonstrate that the pipeline can be executed end to end on real observations.

On this sample we obtain $\hat{\Theta} = 0.20197$. The null-enforcing bootstrap yields $\mathcal{T}_n = 10.2292$, a 95% critical value of 3.6919, and a bootstrap p -value below 0.002, so the FDT null is rejected in this rough public-data proxy exercise. This is economically plausible: the proxy flow is crude and the sample is short, so the result should be read as a proof of concept for the machinery rather than as a market-efficiency verdict.

11.3 Data and Code Availability

Simulation code, plotting scripts, and the evolving `fdt_market` package will be released in a public repository upon completion of the real-data proof-of-concept. Real-data replication files will be shared subject to vendor licensing constraints.

11.4 Acknowledgements, Funding, and Conflicts

The author thanks colleagues and seminar participants for helpful discussions on transient impact, passivity, and high-frequency econometrics. No external funding was received for this project. The author declares no competing interests.

11.5 Supplementary Material

The online supplement will contain additional robustness checks for the parametric bootstrap, extended Monte Carlo tables, and any future long-memory extensions.



Figure 5: Public-data proof of concept for AAPL using one month of 5-minute OHLCV bars and a signed square-root volume proxy for order flow. The dark line is the estimated deviation functional $\hat{\Delta}(\omega)$ across the tested frequency band; the dashed gray baseline marks the FDT null $\Delta(\omega) = 0$.

12 Conclusion

The draft now presents a coherent mode-by-mode derivation of the proposed fluctuation–dissipation relation, a concrete frequency-domain testing recipe, a Monte Carlo study with controlled size and increasing power, a small public-data proof of concept, and a regularized inverse problem for recovering the latent relaxation spectrum. A full TAQ-based empirical section remains the next major step, but the manuscript now demonstrates both simulation evidence and a real-data pipeline prototype.

A Single-Mode Illustration

If $M = A\delta_{\rho_0}$, then

$$\hat{G}(i\omega) = \frac{A\rho_0}{\rho_0 + i\omega}, \quad \text{Im } \hat{G}(i\omega) = \frac{A\omega\rho_0}{\rho_0^2 + \omega^2}.$$

By [theorem 3.1](#), the driving orthogonal increment has spectral density $2\Theta A/\rho_0$. Therefore

$$S_P(\omega) = \left| \frac{\rho_0}{\rho_0 + i\omega} \right|^2 \frac{2\Theta A}{\rho_0} = \frac{2\Theta A\rho_0}{\rho_0^2 + \omega^2} = \frac{2\Theta}{\omega} \text{Im } \hat{G}(i\omega).$$

Finite mixtures

If $M = \sum_{k=1}^K A_k \delta_{\rho_k}$ and the modal noises are orthogonal, then

$$S_P(\omega) = \sum_{k=1}^K \frac{2\Theta A_k \rho_k}{\rho_k^2 + \omega^2} = \frac{2\Theta}{\omega} \sum_{k=1}^K \frac{A_k \omega \rho_k}{\rho_k^2 + \omega^2} = \frac{2\Theta}{\omega} \text{Im } \hat{G}(i\omega).$$

B Symbol Glossary

Symbol	Meaning
$G(t)$	Impact kernel
$\hat{G}(z)$	Fourier–Laplace transform of the impact kernel
$M(d\rho)$	Relaxation measure
$S_P(\omega)$	Price power spectral density
$S_Q(\omega)$	Order-flow spectral density
Θ	Market thermal index
λ	Permanent impact scale
σ_N^2	Noise-trader flow variance rate
$\Delta(\omega)$	Non-equilibrium deviation measure
$\mathcal{T}_n$	Frequency-domain test statistic

C Selected References

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